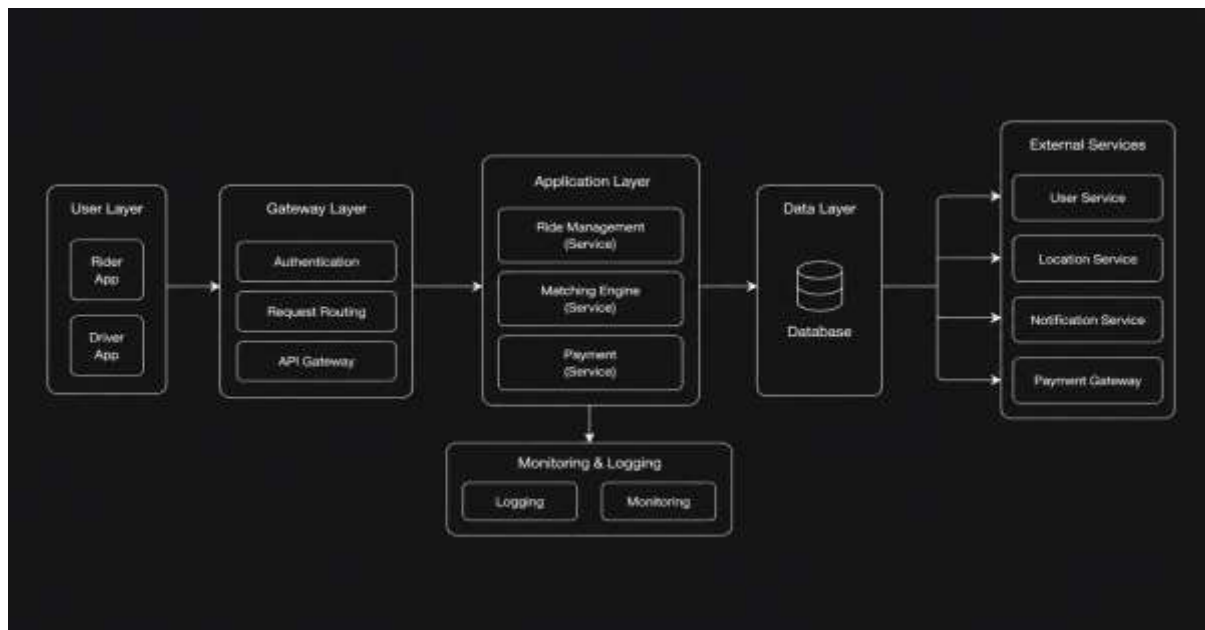


Technical Architecture



1. Client Layer (React.js)

This layer provides the user interface. It allows users to sign in, choose a cab, enter pickup and destination details, and monitor rides through an interactive React.js application.

2. API Layer (Express.js)

The API layer connects the frontend with the backend. Express.js handles client requests through REST APIs such as booking rides, retrieving user details, updating ride information, and cancelling bookings.

3. Service Layer

The service layer contains the main application logic. It computes fares, identifies suitable drivers, manages ride assignments, and updates ride progress based on business rules.

4. Data Access Layer (Mongoose ODM)

This layer interacts with MongoDB using Mongoose. It defines schemas, manages data models, and performs create, read, update, and delete operations efficiently.

Booking Process

A user submits ride details from the React application. The request is sent to the Express API, where the system identifies an available driver, stores the booking in MongoDB, and shares real-time ride updates with both the rider and the driver.